Abdelrahman Rabie Ahmed

Computer and Systems Engineer
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Education

Bachelor of Engineering (B.E.) - Aswan University
Sep 2021 - Jul 2026

DEPT. Computer and Systems Engineering.

- **Graduation Project:** Non-Invasive Blood Glucose Level Estimation Using Smartphone PPG Signals
  - Designed and implemented an end-to-end pipeline for PPG extraction, signal processing, and deep learning-based blood glucose estimation using smartphone camera videos.
- **Graduation Project Grade:** Excellent
- **Cumulative Grade:** Excellent with Honors (Ranked 1st in Class)

Experience

San3a Tech
Apr 2026 – Present

Training Specialist

- Designing and delivering training programs focused on digital fabrication and physical computing to develop learners' technical, practical, and problem-solving skills.

San3a Tech
Apr 2025 – Oct 2025

Junior Specialist Instructor

- Taught digital fabrication and Arduino concepts to empower students in designing and building smart solutions for real-life challenges.
- Mentored a group of 12 girls through hands-on projects, fostering creativity, technical skills, and confidence in prototyping and problem-solving.

Udacity
Aug 2023 – Jul 2025

Coding Instructor

- Taught programming and coding skills to around 300 preparatory and secondary-grade students.
- Developed and delivered engaging, age-appropriate lessons to inspire interest in technology and problem-solving using Python.

Ischool
May 2024 – Aug 2024

Coding Instructor

- Taught programming and coding skills to around 150 primary students using Scratch, Godot, and mBlock.

Faculty of Engineering, Aswan University
Aug 2022 – May 2023

Junior Teaching Assistant

- Assisted students in Introduction to Programming with FORTRAN and C++ to solve basic algorithmic problems.

Trainings and Courses

Embedded Systems Essentials Course
Jul 2025 - Sep 2025

Siemens — *3 Months*

- Embedded architecture fundamentals, including microcontrollers, memory.
- Best practices in agile development and functional safety for embedded applications.

Embedded System Course
Oct 2023 - Nov 2023

TIEC — *15 Days*

- Fundamentals of Embedded Systems, AVR Interfacing using Hardware KIT.

Embedded System Diploma
Aug 2023 - Sep 2023

NTI — *90 Hours*

- C and Embedded C Programming.
- Implemented a system for publishing students grades and inquiring about the results.

Projects

ALU based on FPGA
Dec 2024

Computer Engineering Department, Faculty of Engineering, Aswan University

- Created an ALU based on Basys 3 to perform advanced tasks.

School Degree Management System
Sep 2024 - Dec 2024

Computer Engineering Department, Faculty of Engineering, Aswan University

- Allows School to manage Degrees and print Students Resuls.
- Used OOP in JAVA.

Blood Bank
May 2023

Computer Engineering Department, Faculty of Engineering, Aswan University

- Create a System meet the needs of donors and patients, ensuring safe and efficient blood management.
- Used C++ , OOP and Files

Technical Skills

Hardware Skills

- Development Boards (Arduino, ESP32, Raspberry Pi, micro:bit)
- Embedded C
- Microcontroller Interfacing (AVR)
- Communication Protocols (UART, SPI, I^2C)
- Digital Design (Verilog)
- FPGA (Basys 3)
- Simulation & Design Tools (Proteus, Vivado, Xilinx ISE)

Software Skills

- Programming Languages (C/C++, Java, Python)
- Object-Oriented Programming (OOP)
- Data Structures & Algorithms
- Web Development (HTML, CSS, JavaScript)
- Databases (SQL, MySQL)
- Version Control (Git, GitHub)

Extracurricular Activities

Aswan Practical Engineering Lab - (APEL)

PRESIDENT
Aug 2024 - Jul 2025

- Demonstrated strong leadership, organizational, and teamwork skills as President of a Robotics Student Activity.
- Successfully managed a large scale competition covering Upper Egypt with 42 teams [ARC 9 Time Out].
- Successfully coordinated multiple robotics workshops and organized the team's first-ever media workshop, followed by related competitions.

Technical Head
May 2024 - Aug 2024

- Leading Technical team work in various competitions, and work-shops [ARC8 - phase2.]
- Teaching students about robotics and programming through technical sessions.

Technical Vice
Jun 2023 - May 2024

- Leading Technical team work in various competitions, and work-shops [ARC8 - phase1.]
- Conducted technical sessions on robotics and programming to enhance the skills and efficiency of the team.

Technical Member
Aug 2022 - Jun 2023

- Helping students understand robotics/electronics concepts.
- Member of technical team that succesfully orgranized [ARC7 - phase1 and ARC7 - phase2]

HONORS & AWARDS

- Best Head - APEL · Aug 2024
- 10th Place in Code Storm - IEEE Aswan · Jun 2022
- Best Member - APEL · Jul 2022

Languages

- Arabic
- English